Energy 63 (2013) 233–244

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Time-of-use based electricity demand response for sustainable manufacturing systems

Yong Wang, Lin Li*

Department of Mechanical and Industrial Engineering, University of Illinois at Chicago, Chicago, IL 60607, USA

ARTICLE INFO

Article history:
Received 16 March 2013
Received in revised form
6 September 2013
Accepted 3 October 2013
Available online 31 October 2013

Keywords:
Electricity demand response
Load management
Energy efficiency
Cost effectiveness
Sustainable manufacturing systems

ABSTRACT

As required by the *Energy Policy Act of 2005*, utility companies across the U.S. are offering TOU (time-of-use) based electricity demand response programs. The TOU rate gives consumers opportunities to manage their electricity bill by shifting use from on-peak periods to mid-peak and off-peak periods. Reducing the amount of electricity needed during the peak load times makes it possible for the power grid to meet consumers' needs without building more costly backup infrastructures and help reduce GHG (greenhouse gas) emissions. Previous research on the applications of TOU and other electricity demand response programs has been mainly focused on residential and commercial buildings while largely neglected industrial manufacturing systems. This paper proposes a systems approach for TOU based electricity demand response for sustainable manufacturing systems under the production target constraint. Key features of this approach include: (i) the electricity related costs including both consumption and demand are integrated into production system modeling; (ii) energy-efficient and demand-responsive production scheduling problems are formulated and the solution technique is provided; and (iii) the effects of various factors on the near-optimal scheduling solutions are examined. The research outcome is expected to enhance the energy efficiency, electricity demand responsiveness, and cost effectiveness of modern manufacturing systems.

© 2013 Elsevier Ltd. All rights reserved.

1. Introduction

The industrial sector is the largest energy consumer in the United States [1]. It consumes 31% of the total energy and is responsible for about one third of the total GHG (greenhouse gas) emissions in the country [2,3]. A major portion of the energy consumed by the industrial sector is in the form of electricity [1]. Electricity is a form of energy that cannot be effectively stored in bulk. It must be generated, distributed, and consumed immediately. Consumers' needs change vastly in different seasons and even at different time of a day [4]. In order to meet the needs during peak periods, a huge array of expensive equipment including generators, transformers, wires, and substations has to be kept on constant standby, otherwise the system will become unstable and blackouts may occur. This requires large extra investments for those backup infrastructures. By 2030, about $2 trillion investments for new generation capacities, transmission, and distribution will be required to satisfy the growing needs [5]. On average, 1 kW hour (kWh) of electricity generation causes 1.56 pounds (0.71 kg) of GHG emissions [6]. Backup generators are often dirtier and less efficient than base load generators, and therefore create more GHG emissions for each kWh of electricity generated.

GHG emissions have become a vital issue to the sustainable development of human society since they are recognized as the leading cause of global warming and climate change. Under the increasingly rising pressures of reducing GHG emissions from both domestic and international society, many regulations have been enacted to curb the emissions. The U.S. government has set up a target to reduce energy use to 17% below 2005 levels by 2020 [7]. Accordingly, technologies that may promisingly reduce GHG emissions and postponing or eliminating the huge extra investments have attracted great public interests. One such technology is demand response.

The U.S. *FERC (Federal Energy Regulatory Commission)* [8] defines demand response as *"changes in electric use by demand-side resources from their normal consumption patterns in response to changes in the price of electricity, or to incentive payments designed to induce lower electricity use at times of high wholesale market prices or when system reliability is jeopardized."* Demand response targets at reducing peak demand to control the risk of potential disturbances, avoiding extra investments in additional infrastructures, avoiding use of more expensive and less efficient generators, and thus cutting GHG emissions. It is estimated that the implementation of

* Corresponding author. Tel.: +1 312 9963045; fax: +1 312 4130447.
E-mail address: linli@uic.edu (L. Li).

0360-5442/$ – see front matter © 2013 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.energy.2013.10.011

Nomenclature

Bold face

$\boldsymbol{S}$	position matrix of the binary PSO algorithm
$\boldsymbol{S}_{PB}$	personal best position matrix of the binary PSO algorithm
$\boldsymbol{S}_{GB}$	global best position matrix of the binary PSO algorithm
$\boldsymbol{V}$	velocity matrix of the binary PSO algorithm
$\boldsymbol{X}(t)$	a column vector containing the state probabilities of the system at the end of time slot t

Upper case

$BL_i(t)$	blockage probability of machine m_i during time slot t
C_i	capacity of buffer b_i (the largest number of parts the buffer can hold)
CP	average cumulative production of the system during the planning horizon
CPo	target cumulative production of the system during the planning horizon
H	number of hours in the finite planning horizon
N	number of machines in the manufacturing system
N_P	swarm size of the binary PSO algorithm
N_T	iteration number of the binary PSO algorithm
$PR_i(t)$	production rate of machine m_i ($i = 1$ or N) during time slot t
$PR_{SYS}(t)$	production rate of the system during time slot t
$Q_{i,(j_2\|j_1)}(t)$	transition probability from state j_1 to j_2 during time slot t for buffer b_i
$ST_i(t)$	starvation probability of machine m_i during time slot t
T	number of total time slots during the planning horizon
$WIP_i(t)$	work-in-process inventory of buffer b_i at the end of time slot t
$WIP_{SYS}(t)$	total work-in-process of the system at the end of time slot t

Lower case

b_i	index of the ith buffer in the manufacturing system, $i = 1, ..., N-1$
$b(t)$	billable cost indicator
c_{DT}	cost of the billable power demand of the system during the planning horizon
$c_D(t)$	TOU demand rate (\$/kW) during time slot t
c_{ET}	cost of the total electricity consumption of the system during the planning horizon
$c_E(t)$	TOU consumption rate (\$/kWh) during time slot t
c_{Fixed}	fixed charge during the planning horizon
c_T	total electricity cost
d_i	electric power (in kW) drawn by machine m_i when it is up
$d_{SYS}(t)$	power demand of the system during time slot t
d_T	billable power demand of the system (the highest average kW measured in any on-peak 15-minute interval during the planning horizon)
e_i	electric energy (in kWh) consumed by machine m_i when it is up
$e_{SYS}(t)$	electricity consumption of the system during time slot t
e_T	total electricity consumption of the system during the planning horizon
i, j, j_1, j_2, k, t	general indexes
t_C	cycle time (the time needed by a machine to process a part)
l	ceiling integer number of the time slots in any 15-minute interval
m_i	index of the ith machine in the manufacturing system, $i = 1, ..., N$
p_i'	reliability of machine m_i
$p_i(t)$	probability of machine m_i being up during time slot t (considering both machine reliability and control signal)
$q_{i,j}(t)$	probability of buffer b_i in state j ($j = 0, \ldots, C_i$) at the end of time slot t
$s_i(t)$	scheduled control signal ("on" or "off") for machine m_i during time slot t

Greek

$\theta_0, \theta_1, \theta_2$	parameters of the binary PSO algorithm

Functions

$G(\cdot)$	state transition dynamics function
$\text{rand}(\cdot, \cdot)$	uniformly distributed random number generator

demand response programs together with energy efficiency improvement can reduce the needs for new generation capacities from 214 GW to 133 GW in 2030, by 38% [5]. Recent research results have also suggested that demand response can be used as a solution to meet supply-demand fluctuations in the grid with significant penetration of variable renewable energy sources of intermittent nature [9–12].

The term demand response encompasses a wide range of solutions and mechanisms. According to the *2012 Survey on Demand Response and Advanced Metering* by U.S. FERC [8], TOU (time-of-use) pricing is among the most popular demand response mechanisms. It utilizes time sensitive pricing structures to spread the costs of the needs for extra equipment. The mechanism encourages the electricity consumers to shift their power demand from peak periods (with high prices) to off-peak periods (with low prices). TOU pricing is widely available from utility companies across the U.S. thanks to the *Energy Policy Act of 2005* [13]. There are about 150 entities providing different sorts of TOU pricing programs. These entities represent all aspects of the electricity delivery industry: investor-owned utilities, municipal utilities, rural electric cooperatives, power marketers, state and federal agencies, and other demand response providers. A full list of the entities' names is available in the appendix of the survey report [8]. TOU pricing is also one of the easiest implementations of demand response due to less stringent technological requirements.

Most TOU pricing profiles, like the one provided in Table 1 [14], divide the day into two or three periods and assign prices for each period [15]. Electricity consumption and demand are tracked by smart meters [16–18] and both components count towards consumers' monthly bill. The consumption rate is formulated in

Table 1
A representative TOU pricing profile [14].

Season	Type	Time of day	Consumption rate (\$/kWh)	Demand rate (\$/kW)	Fixed charge (\$)
Summer	Off-peak	7pm–1pm	0.08274	0	51.42
(Jun–Sep)	On-peak	1pm–7pm	0.16790	18.80	
Winter	Off-peak	9pm–10am	0.08274	0	
(Oct–May)	On-peak	10am–9pm	0.11224	8.12	

$/kWh. The charge for consumption is determined based on the time-of-day kWh readings during the monthly billing period. The demand rate is formulated in $/kW. The charge for demand is determined based on the highest average kW measured in any on-peak 15-minute interval during the monthly billing period. In addition, there is a separate fixed charge, which generally includes the customer charge and metering charges during the monthly billing period. A consumer's total bill is rendered by adding up the subtotals of all the three parts.

Existing applications of the TOU and other demand response programs are predominantly limited to the residential and commercial building sectors [19–29]. Many research results have provided the guidance in various aspects for the consumers so that they can manage the electricity consumed by lighting, HVAC, kitchen appliances, and laundry equipment in response to the varying prices. Both manual and automatic control strategies have been extensively studied to reduce the electricity demand of buildings during peak periods.

By contrast, the research progress of the TOU and other demand response programs for the industrial sector lags far behind. Existing efforts in this area focus only on isolated or mutually independent machines or processes [30–33]. These efforts have largely ignored the following properties shard in many industrial systems of modern manufacturing [34,35]: (i) the system generally consists of multiple machines and buffers; (ii) machines are unreliable and they are subject to unexpected failures; (iii) the use of buffers between machines may mitigate the negative effects of machine downtime; and (iv) the system exhibits highly complex dynamics because each machine's operation states are determined not only by the machine itself, but also by other machines and buffers. Therefore, it is more important and meaningful to consider the interactions within the manufacturing systems for effective demand response. Manufacturing system modeling considering the interactions is an active research area [36–43]. Although a few publications [44,45] related to the demand response technology are available in this area, none of them explicitly consider practical issues such as limited buffer capacities as well as machine breakdown, starvation, and blockage. Consequently, the claimed benefits of the demand response actions may be unachievable.

Motivated by the status quo, this paper proposes to research on the TOU based production scheduling that can help manufacturers manage electricity costs during peak periods. A time-dependent method for modeling typical manufacturing systems will be developed to involve control actions for optimal demand response. TOU pricing is especially suitable for such application in the manufacturing industry. When the pricing profile is provided, manufacturers can make decisions by jointly considering the production target together with electricity consumption and demand. This is made possible by the utilization of buffers in the system that allow for temporary stoppage of work in one area without affecting the entire system throughput. Therefore, a schedule can be created to control the status of each machine to minimize concurrent operations of all the machines during peak hours. The appropriate implementation of this research outcome may lead to energy-efficient, demand-responsive, and cost-effective operations of modern manufacturing systems.

The rest of this paper is organized as follows. Section 2 models the dynamics and performance measures of general manufacturing systems with multiple machines and buffers. Electricity related costs based on the TOU pricing profile including both electricity consumption and power demand are also integrated in the model. Production scheduling problem formulations and the solution technique are discussed in Section 3. Effects of various factors on the TOU based scheduling solutions are examined in Section 4. Conclusions are drawn in Section 5.

2. Manufacturing system modeling integrating electricity consumption and demand

Manufacturing system modeling can generate valuable information that is helpful for production cost analysis, optimal system design, control, and management. The complicated interrelations between component-level characteristics (e.g., machine reliabilities, cycle time, and buffer capacities) and system-level characteristics (e.g., production rate, electricity consumption and demand) will be investigated in this section.

2.1. Conventions

A typical manufacturing system is shown in Fig. 1. The system consists of N machines (denoted by squares) and $N-1$ buffers (denoted by circles) connected in series. Suppose we consider a finite planning horizon of H hours, which is evenly discretized into T slots, with $t = 1$ being the first slot, and $t = T$ being the last. The slot duration is equal to the cycle time t_C of the machines, i.e., $t_C = H/T$. The cycle time represents the time needed by a machine to process a part. Due to internal failure, machine m_i ($i = 1, ..., N$) is up during time slot t with probability p_i' and down with probability $1 - p_i'$. Let $s_i(t)$ be the status of the scheduled control signal for machine m_i during time slot t. It is defined as

$$s_i(t) = \begin{cases} 1, & \text{if the control signal is set to "on"} \\ 0, & \text{if the control signal is set to "off"} \end{cases} \quad (1)$$

Therefore, considering the machine reliability and the control signal together, machine m_i is up during time slot t with probability $p_i(t) = s_i(t)p_i'$ and down with probability $1 - p_i(t) = 1 - s_i(t)p_i'$.

For machine m_i, when it is up during time slot t, it consumes e_i kWh of electric energy and draws d_i kW of electric power; when it is down, it consumes no energy or power. Let the capacity of buffer b_i ($i = 1, ..., N-1$) be C_i, which is the largest number of parts the buffer can hold. The blocked-before-service and time-dependent-failure conventions are adopted. The first machine is never starved; the last machine is never blocked. Buffer states are defined by the number of parts it contains at the end of a time slot. Machine states are defined by its working status such as producing, starved, blocked, and down during a time slot. Each machine's uptime and downtime are determined independently from the other machines'. Buffer b_i's state (occupancy) can change in each time slot at most by one part.

2.2. Modeling of system dynamics and performance measures

This paper deals with time-dependent up probability $p_i(t)$ instead of only a fixed p_i'. The derivations of some most important equations are given in the appendix section.

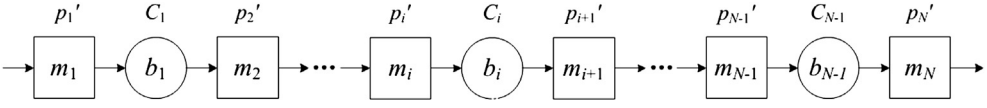


Fig. 1. A typical manufacturing system.

Let $Q_{i,(j_2|j_1)}(t)$ be the transition probability from state j_1 to j_2 during time slot t for buffer b_i ($i = 1, \ldots, N-1$). Let $ST_i(t)$ be the starvation probability of machine m_i ($i = 1, \ldots, N$) during time slot t, and let $BL_i(t)$ be the blockage probability of machine m_i during time slot t. Since each buffer's state (occupancy) can change in each time slot at most by one part, the transition matrix of buffer b_i during time slot t can be represented by

$$\mathbf{Q}_i(t) = \begin{bmatrix} Q_{i,(0|0)}(t) & Q_{i,(0|1)}(t) & & & & & & \\ Q_{i,(1|0)}(t) & Q_{i,(1|1)}(t) & \ddots & & & & & \\ & \ddots & \ddots & \ddots & & & & \\ & & \ddots & Q_{i,(j|j)}(t) & Q_{i,(j|j+1)}(t) & & & \\ & & & Q_{i,(j+1|j)}(t) & Q_{i,(j+1|j+1)}(t) & \ddots & & \\ & & & & \ddots & \ddots & & \\ & & & & & \ddots & Q_{i,(C_i-1|C_i-1)}(t) & Q_{i,(C_i-1|C_i)}(t) \\ & & & & & & Q_{i,(C_i|C_i-1)}(t) & Q_{i,(C_i|C_i)}(t) \end{bmatrix} \tag{2}$$

where

$$\begin{aligned}
&Q_{i,(0|0)}(t) = 1 - p_i(t) + ST_i(t), \quad (C_i \geq 1)\\
&Q_{i,(0|1)}(t) = [1 - p_i(t) + ST_i(t)][p_{i+1}(t) - BL_{i+1}(t)], \quad (C_i \geq 1)\\
&Q_{i,(1|0)}(t) = p_i(t) - ST_i(t) \quad (C_i \geq 1)\\
&Q_{i,(j|j)}(t) = [p_i(t) - ST_i(t)][p_{i+1}(t) - BL_{i+1}(t)] + [1 - p_i(t) + ST_i(t)][1 - p_{i+1}(t) + BL_{i+1}(t)], \quad (j = 1, \ldots, C_i - 1;\ C_i \geq 2)\\
&Q_{i,(j|j+1)}(t) = [1 - p_i(t) + ST_i(t)][p_{i+1}(t) - BL_{i+1}(t)], \quad (j = 1, \ldots, C_i - 1;\ C_i \geq 2)\\
&Q_{i,(j+1|j)}(t) = [p_i(t) - ST_i(t)][1 - p_{i+1}(t) + BL_{i+1}(t)], \quad (j = 1, \ldots, C_i - 1;\ C_i \geq 2)\\
&Q_{i,(C_i|C_i)}(t) = [p_i(t) - ST_i(t)][p_{i+1}(t) - BL_{i+1}(t)] + [1 - p_{i+1}(t) + BL_{i+1}(t)], \quad (C_i \geq 1)
\end{aligned} \tag{3}$$

Let $q_{i,j}(t)$ be the probability of buffer b_i ($i = 1, \ldots, N-1$) in state j ($j = 0, \ldots, C_i$) at the end of time slot t. According to the law of total probability, the evolution of $q_{i,j}(t)$ can be described by the following constrained non-linear dynamical system:

$$q_{i,j}(t) = \sum_{k=0}^{C_i} Q_{i,(j|k)}(t) q_{i,k}(t-1), \quad (j = 0, \ldots, C_i) \tag{4}$$

which is subject to

$$\sum_{j=0}^{C_i} q_{i,j}(t-1) = 1 \tag{5}$$

After we substitute the derived equations in (3) for the transition probabilities $Q_{i,(j|k)}(t)$ in (4), the dynamical system can be re-written as

$$\begin{aligned}
&q_{i,0}(t) = [1-p_i(t)+ST_i(t)]q_{i,0}(t-1)+[1-p_i(t)+ST_i(t)][p_{i+1}(t)-BL_{i+1}(t)]q_{i,1}(t-1), \ (C_i \geq 1)\\
&q_{i,1}(t) = [p_i(t)-ST_i(t)]q_{i,0}(t-1)+\{[p_i(t)-ST_i(t)][p_{i+1}(t)-BL_{i+1}(t)]+[1-p_i(t)+ST_i(t)][1-p_{i+1}(t)+BL_{i+1}(t)]\}q_{i,1}(t-1)\\
&\qquad +[1-p_i(t)+ST_i(t)][p_{i+1}(t)-BL_{i+1}(t)]q_{i,2}(t-1), \ (C_i \geq 2)\\
&q_{i,j}(t) = [p_i(t)-ST_i(t)][1-p_{i+1}(t)+BL_{i+1}(t)]q_{i,j-1}(t-1)+\{[p_i(t)-ST_i(t)][p_{i+1}(t)-BL_{i+1}(t)]+[1-p_i(t)+ST_i(t)][1-p_{i+1}(t)+BL_{i+1}(t)]\}\\
&\qquad q_{i,j}(t-1)+[1-p_i(t)+ST_i(t)][p_{i+1}(t)-BL_{i+1}(t)]q_{i,j+1}(t-1), \ (j=2,\ldots,C_i-1; C_i \geq 3)\\
&q_{i,C_i}(t) = [p_i(t)-ST_i(t)][1-p_{i+1}(t)+BL_{i+1}(t)]q_{i,C_i-1}(t-1)+\{[p_i(t)-ST_i(t)][p_{i+1}(t)-BL_{i+1}(t)]+[1-p_{i+1}(t)+BL_{i+1}(t)]\}q_{i,C_i}(t-1), \ (C_i \geq 2)\\
&q_{i,C_i}(t) = [p_i(t)-ST_i(t)]q_{i,C_i-1}(t-1)+\{[p_i(t)-ST_i(t)][p_{i+1}(t)-BL_{i+1}(t)]+[1-p_{i+1}(t)+BL_{i+1}(t)]\}q_{i,C_i}(t-1), \ (C_i = 1)
\end{aligned} \tag{6}$$

where

$$ST_i(t) = \begin{cases} 0, & \text{if } i = 1 \\ p_i(t) q_{i-1,0}(t-1), & \text{if } i = 2, \ldots, N \end{cases} \tag{7}$$

and

$$BL_i(t) = \begin{cases} p_i(t) q_{i,C_i}(t-1)[1-p_{i+1}(t)+BL_{i+1}(t)], & \text{if } i = 1, \ldots, N-1 \\ 0, & \text{if } i = N \end{cases} \tag{8}$$

Define a column vector for time slot t

$$\mathbf{X}(t) \equiv \Big[q_{1,0}(t), \ldots, q_{1,C_1}(t), \ldots\ldots, q_{i,0}(t), \ldots, q_{i,C_i}(t), \ldots\ldots, q_{N-1,0}(t), \ldots, q_{N-1,C_{N-1}}(t)\Big]^{\mathrm{T}} \tag{9}$$

After substituting the equations (7) and (8) for $ST_i(t)$ and $BL_i(t)$ in (6), the dynamics of the system can be formulated in the form

$$\mathbf{X}(t) = G[\mathbf{X}(t-1)] \tag{10}$$

where $\boldsymbol{X}(t)$ contains the state probabilities of the system at the end of time slot t, $\boldsymbol{X}(t-1)$ contains the state probabilities of the system at the end of time slot $t-1$, and function $G(\cdot)$ represents the state

transition dynamics. Therefore, $\boldsymbol{X}(t)$ together with $q_{i,j}(t)$ ($i = 1, \ldots, N-1$; $j = 0, \ldots, C_i$), $\mathrm{ST}_i(t)$, and $\mathrm{BL}_i(t)$ can be obtained iteratively with given initial conditions $\boldsymbol{X}(0)$. For example, if the system starts with all the buffers being empty, which is referred to as a "fresh start" in this paper, then

$$\mathbf{X}(0) = \begin{bmatrix} \mathbf{q}_1(0) \\ \vdots \\ \mathbf{q}_i(0) \\ \vdots \\ \mathbf{q}_{N-1}(0) \end{bmatrix} \tag{11}$$

where

$$\mathbf{q}_i(0) = \left[q_{i,0}(0), q_{i,1}(0), \ldots, q_{i,C_i}(0)\right]^{\mathrm{T}} = [1, 0, \ldots, 0]^{\mathrm{T}}, \quad i = 1, \ldots, N-1 \tag{12}$$

After $\boldsymbol{X}(t)$ is obtained, its elements can be used to derive the following system performance measures. The work-in-process inventory of buffer b_i ($i = 1, \ldots, N-1$) at the end of time slot t is

$$\mathrm{WIP}_i(t) = \sum_{j=0}^{C_i} j q_{i,j}(t) \tag{13}$$

The total work-in-process of the system at the end of time slot t is

$$\mathrm{WIP}_{\mathrm{SYS}}(t) = \sum_{i=1}^{N-1} \mathrm{WIP}_i(t) \tag{14}$$

The production rate of machine m_i ($i = 1$ or N) during time slot t is

$$\mathrm{PR}_i(t) = p_i(t) - \mathrm{ST}_i(t) - \mathrm{BL}_i(t) \tag{15}$$

The production rate of the system during time slot t is equal to the production rate of the last machine m_{N}, i.e.,

$$\mathrm{PR}_{\mathrm{SYS}}(t) = \mathrm{PR}_N(t) \tag{16}$$

The average cumulative production of the system during the planning horizon of T time slots is

$$\mathrm{CP} = \sum_{t=1}^{T} \mathrm{PR}_{\mathrm{SYS}}(t) \tag{17}$$

2.3. Modeling of electricity consumption and power demand

The electricity consumption of the N-machine system shown in Fig. 1 during time slot t is

$$e_{\mathrm{SYS}}(t) = \sum_{i=1}^{N} e_i p_i(t) = \sum_{i=1}^{N} e_i p_i' s_i(t) \tag{18}$$

The power demand of the system during time slot t is

$$d_{\mathrm{SYS}}(t) = \sum_{i=1}^{N} d_i s_i(t) \tag{19}$$

The total electricity consumption of the system during the planning horizon is

$$e_{\mathrm{T}} = \sum_{t=1}^{T} e_{\mathrm{SYS}}(t) \tag{20}$$

Let l be the ceiling integer number of the time slots in any 15-minute interval, i.e.,

$$l = \left\lceil \frac{15/60}{t_{\mathrm{C}}} \right\rceil \tag{21}$$

The billable power demand of the system, which is defined as the highest average kW measured in any on-peak 15-minute interval during the planning horizon, is

$$d_{\mathrm{T}} = \max_t \frac{\sum_{t_1=t}^{t+l-1} d_{\mathrm{SYS}}(t_1) b(t_1)}{l} \tag{22}$$

where

$$b(t) = \begin{cases} 1, & \text{if the TOU demand rate for time slot } t \text{ is larger than 0} \\ 0, & \text{if the TOU demand rate for time slot } t \text{ is 0} \end{cases} \tag{23}$$

2.4. TOU cost analysis of electricity consumption and power demand

For the manufacturing system shown in Fig. 1, the cost of the total electricity consumption of the system during the planning horizon is

$$c_{\mathrm{ET}} = \sum_{t=1}^{T} e_{\mathrm{SYS}}(t) c_{\mathrm{E}}(t) \tag{24}$$

where $c_{\mathrm{E}}(t)$ is the TOU consumption rate (\$/kWh) during time slot t. The cost of the power demand of the system during the planning horizon is

$$c_{\mathrm{DT}} = \max_t \frac{\sum_{t_1=t}^{t+l-1} d_{\mathrm{SYS}}(t_1) c_{\mathrm{D}}(t_1)}{l} \tag{25}$$

where $c_{\mathrm{D}}(t_1)$ is the TOU demand rate (\$/kW) during time slot t_1. The total electricity cost can be formulated as

$$c_{\mathrm{T}} = c_{\mathrm{DT}} + c_{\mathrm{ET}} + c_{\mathrm{Fixed}} \tag{26}$$

where c_{Fixed} is a fixed charge during the planning horizon.

The established models (system dynamics, performance measures, electricity consumption, and power demand) as well as the cost analysis are essential for creating appropriate production schedules for effective electricity demand response in Section 3.

3. Production scheduling for TOU based demand response

Based on the knowledge obtained in Section 2, we will be able to make decisions on production schedules by jointly considering the production target and electricity costs.

3.1. Production scheduling problem formulations

For the manufacturing system with N machines and $N-1$ buffers, a production schedule can be created to control at what time, which machine(s) can be shut down temporarily to lower electricity consumption and power demand as long as an average

cumulative production target, i.e., CP_0 products, is achieved for the planning horizon. We choose the control signal $s_i(t)$ of each machine m_i ($i = 1, \ldots, N$) to be the decision variables, the following two production scheduling problems with different objectives can be formulated.

Formulation 1: Minimizing the total electricity consumption while maintaining an amount of average cumulative production that is not lower than the required level, i.e.,

$$\min_{s_i(t)} \; e_T, \quad \text{subject to}: \; CP \geq CP_0 \tag{27}$$

Formulation 2: Minimizing the total electricity cost while maintaining an amount of average cumulative production that is not lower than the required level, i.e.,

$$\min_{s_i(t)} \; c_T, \quad \text{subject to}: \; CP \geq CP_0 \tag{28}$$

Solving these two problems will provide optimal schedules of control signals for setting each machine in the "on" or "off" status to achieve corresponding objectives.

3.2. Meta-heuristic techniques for near-optimal solutions

The formulated scheduling problems present a huge challenge with regard to finding the optimal solutions. The calculations of electricity consumption e_T and total electricity cost c_T are pretty straightforward as shown in Sections 2.3 and 2.4. However, the calculation of the average cumulative production CP in Section 2.2 is highly non-linear because it relies on iterative operations to untangle the complex interactions among all the machines and buffers. Besides, the decision variable $s_i(t)$ is binary and not continuous. The formulations are both ZONLP (zero-one nonlinear programing) problems due to these factors.

ZONLP is a special case of MINLP (mixed integer nonlinear programing). Due to the curse of dimensionality, there are still no exact methods that can effectively solve the general MINLP problems. Recently, meta-heuristic methods such as GAs (genetic algorithms), PSO (particle swarm optimization), and TS (tabu search) have gained popularity in both academia and industrial applications. These methods impose no restrictions such as convexity, continuity, and differentiability on the optimization problems. Among these methods, PSO has been widely studied [46–53]. For these reasons, we propose to search for near-optimal solutions to the formulated scheduling problems using a classic binary version of PSO [54].

The original binary PSO algorithm is slightly tailored for this research work. The implementation is briefly described in this section. The algorithm randomly generates a large group of members named particles. Each particle is characterized by a position matrix $\boldsymbol{S}$ and a velocity matrix $\boldsymbol{V}$. The position matrix $\boldsymbol{S}$ represents a potential solution to Formulation 1 or 2 in the encoded form

$$\mathbf{S} \equiv \begin{bmatrix} \boldsymbol{S}_1(1) & \boldsymbol{S}_1(2) & \cdots & \boldsymbol{S}_1(T) \\ \boldsymbol{S}_2(1) & \boldsymbol{S}_2(2) & \cdots & \boldsymbol{S}_2(T) \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{S}_N(1) & \boldsymbol{S}_N(2) & \cdots & \boldsymbol{S}_N(T) \end{bmatrix} \tag{29}$$

The velocity matrix $\boldsymbol{V}$ is of the same size as $\boldsymbol{S}$, and it modifies the position matrix iteratively during the search process.

Before the search process begins, the position matrixes and the velocity matrixes of the entire swarm are initialized so that the elements in $\boldsymbol{S}$'s and $\boldsymbol{V}$'s are assigned random integer numbers drawn from the set $\{-1, 0, 1\}$. During each search iteration, the position and the velocity of each particle are updated by the following element-by-element operations

$$\begin{aligned} \mathbf{V}(i,j) \; = \; & \theta_0 \cdot \mathbf{V}(i,j) + \text{rand}(0,\theta_1) \cdot [\mathbf{S}_{PB}(i,j) - \mathbf{S}(i,j)] \\ & + \text{rand}(0,\theta_2) \cdot [\mathbf{S}_{GB}(i,j) - \mathbf{S}(i,j)] \end{aligned} \tag{30}$$

$$\mathbf{S}(i,j) \; = \; \begin{cases} 1, & \text{if rand}(0,1) < \frac{1}{1+e^{-|\mathbf{S}(i,j)+\mathbf{V}(i,j)|}} \\ 0, & \text{otherwise} \end{cases} \tag{31}$$

In (30) and (31), (i, j) means the element in the ith row and jth column in the matrix; rand$(0, \theta_1)$ and rand$(0, \theta_2)$ generate random numbers from the uniform distribution in the range $[0, \theta_1]$ and $[0, \theta_2]$, respectively; $\boldsymbol{S}_{PB}$ is the personal best solution that the particle has found. $\boldsymbol{S}_{GB}$ is the global best solution of the entire swarm; θ_0 is a scalar inertial weight. A large inertia weight facilitates the search of new areas, while a small inertia weight facilitates finer search of the currently focused promising area. Careful selection of parameters θ_0, θ_1, and θ_2 can properly balance between exploration and exploitation to avoid premature convergence to a local optimum yet still ensure a good rate of convergence to the optimum.

The swarm size (i.e., the number of particles in the swarm) is denoted with N_P. The algorithm terminates after a certain number of PSO iteration N_T. The flowchart of the binary PSO algorithm is illustrated in Fig. 2.

More details on the near-optimal PSO solutions to the scheduling formulations will be discussed in Section 4.

4. Case studies and effects of various factors on the TOU based scheduling solutions

The problem formulations, the TOU pricing profile, and the manufacturing system parameters may all significantly influence the scheduling solutions, and hence the effectiveness and attractiveness of the TOU based demand response program to consumers. Therefore, the effects of these factors should be carefully investigated.

4.1. Insights gained from an illustrative case

As an example, we consider a manufacturing system with $N = 3$ machines and 2 buffers. Suppose the planning horizon is one working day, which consists of two 8-hour shifts (from 8 a.m. to 12 midnight), i.e., $H = 16$ h. Suppose the cycle time $t_C = 0.25$ h (i.e., 15 min) for each machine. Thus, the planning horizon is divided into $T = H/t_C = 64$ slots. The following parameters of the machines

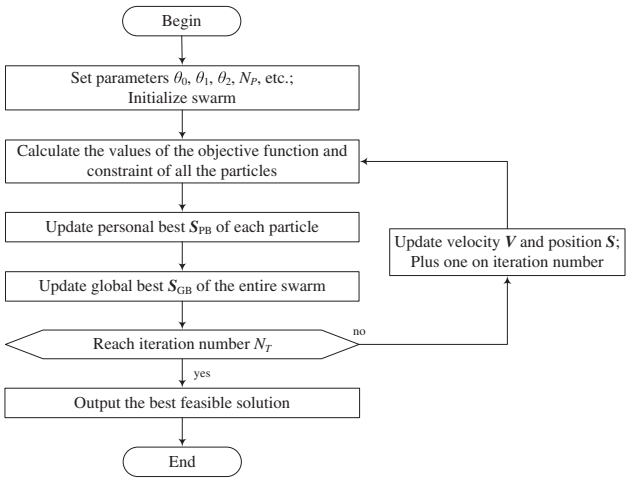


Fig. 2. Binary PSO algorithm.

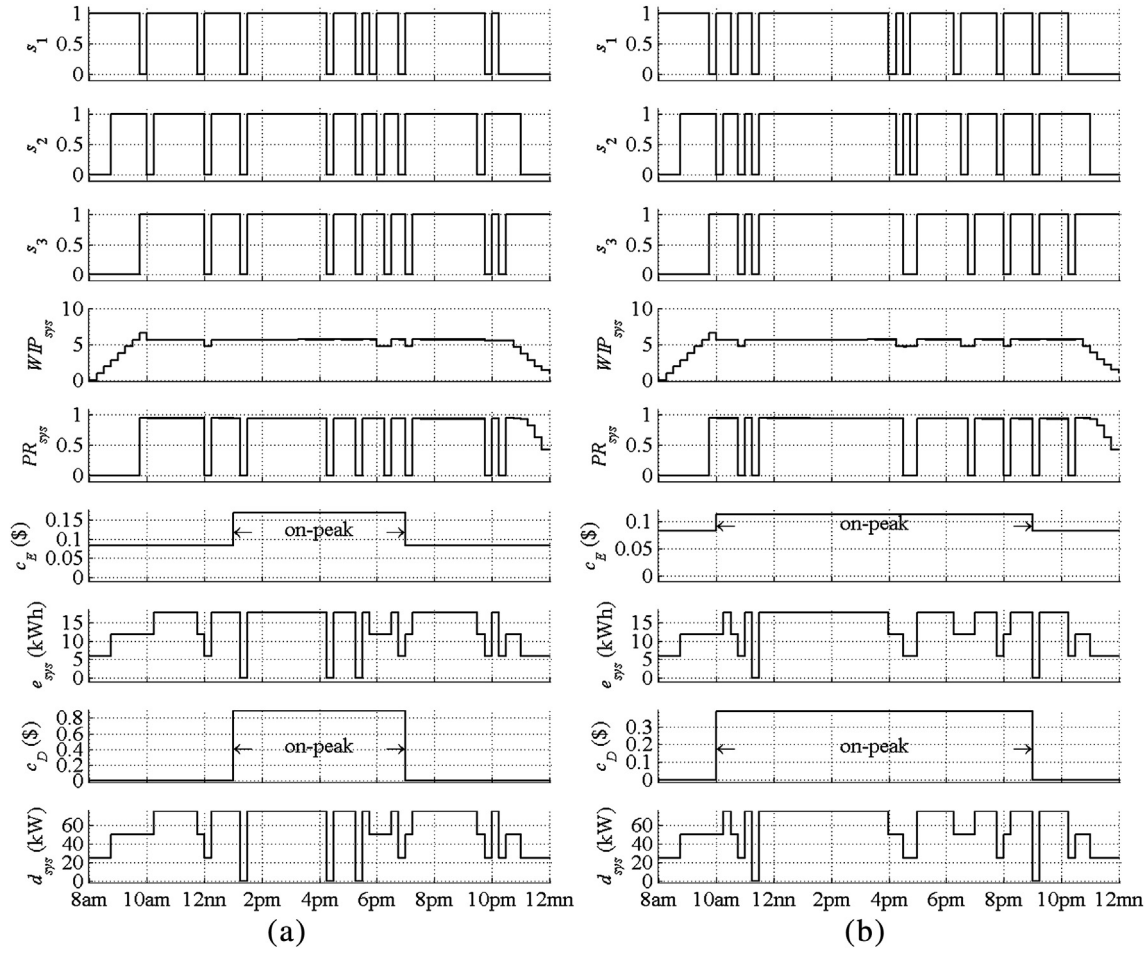


Fig. 3. Best schedules for Formulation 1 based on (a) the summer and (b) the winter TOU rates.

and buffers are given: machine electric power $d_1 = d_2 = d_3 = 25$ kW; machine electric energy consumption during time slot t when up $e_1 = e_2 = e_3 = 6.25$ kWh; machine reliabilities $p'_1 = p'_2 = p'_3 = 0.95$; and buffer capacities $C_1 = C_2 = 5$. The parameters are set to be identical for illustration purposes. However, the applicability of the approach proposed in this paper is not limited to manufacturing systems with identical machine or buffer parameters. The task is to find near-optimal production schedules regarding Formulations 1 and 2, respectively. The schedules will be repeated every working day with targeted average cumulative production of $CP_0 = 45$ products following a fresh start. The TOU consumption rates $c_E(t)$ for different time periods are provided in Table 1. The TOU demand rates $c_D(t)$ and the fixed charge c_{Fixed} are also provided in Table 1 but they should be divided by 21, the number of working days in a month, to obtain the daily equivalents.

Parameter selection research for the PSO algorithm has been previously conducted in [55–59]. In this paper we set swarm size $N_P = 1000$ and iteration number $N_T = 2000$. The parameters $\theta_0 = 1$, $\theta_1 = 2$, and $\theta_2 = 2$ reported in Ref. [59] perform generally well and they are also adopted in the implemented binary PSO algorithm.

For Formulation 1, we run 20 trials of the binary PSO program first based on the summer TOU pricing profile. The time consumption of the implemented binary PSO algorithm for each trial is approximately 5 s on a computer equipped with a processor of Intel (R) Xeon(R) CPU (W3550 @ 3.07 GHz 3.06 GHz) and a memory of 4 GB. Multiple best solutions have been obtained. Although the exact schedules for these solutions are different, they all correspond to the same minimal e_T out of the 20 trials. We randomly pick one best solution $s_i(t)$ $(i = 1, 2, 3)$ and plot it in Fig. 3 (a). Then we run 20 trials of the binary PSO program based on the winter TOU pricing profile and plot the best solution in Fig. 3 (b). Also plotted are the trajectories of some important measures over time t. These measures are $WIP_{SYS}(t)$, $PR_{SYS}(t)$, $e_{SYS}(t)$, and $d_{SYS}(t)$. The pre-specified TOU consumption and demand rates $c_E(t)$ and $c_D(t)$ are shown for reference only. The detailed information of the best summer and winter schedules for Formulation 1 is provided in Table 2. The objective and constraint values corresponding to the schedules for Formulation 1 are listed in Table 3. These results can be explained as follows:

(i) The control signals for the three machines are represented by s_1, s_2, and s_3. These signals are all changing over time. Their changing leads to the changing of the expected values of the performance measures WIP_{SYS}, PR_{SYS}, e_{SYS}, and d_{SYS}.
(ii) At the beginning (about 8am) of the production, the three machines are turned "on" (i.e., $s_i = 1$) sequentially. The first one is turned on immediately, while the second and third ones are turned on with gradually increasing delays. This schedule is meant to conserve electricity on the second and third machines, because the first and the second buffers are all empty after a fresh start and some time is needed for the inventory (i.e., WIP_{SYS}) to build up.
(iii) At the end (about 12 midnight) of the production, the three machines are turned "off" (i.e., $s_i = 0$) sequentially. The first

Table 2
Best schedules for Formulation 1.

Formulation	Solutions $s_i(t)$
Formulation 1 (summer)	$s_1(t)$ = [1111111011111110111110111111111110111010111011111111111010000000]
	$s_2(t)$ = [0001111101111111011110111111111110111011011011111111110111110000]
	$s_3(t)$ = [0000000111111111011101111111111101110111011011111111110101111111]
Formulation 1 (winter)	$s_1(t)$ = [1111111011011011111111111111111110101111110111110111011110000000]
	$s_2(t)$ = [0001111101101011111111111111111110101111101111011101111110000000]
	$s_3(t)$ = [0000000111101011111111111111111111001111111011110111011110111111]

Table 3
Objective and constraint values corresponding to the best schedules for Formulation 1.

Formulation	e_T (kWh)	d_T (kW)	c_{ET} ($)	c_{DT} ($)	c_T ($)	CP
Formulation 1 (summer)	872.813	75	103.992	67.143	171.135	45.217
Formulation 1 (winter)	872.813	75	94.458	29.000	123.458	45.217

one is turned off first, while the second and third ones are turned off with gradually increasing delays. This schedule is meant to conserve electricity on the first and second machines, because the built-up inventory (i.e., WIP_{SYS}) in the first and the second buffers takes some time to be depleted.

(iv) The on-peak period in the summer season is much short than the on-peak period in the winter season. This is represented in the plots of c_E and c_D. Other than the beginning and ending periods of the production, the machines are controlled so that the energy consumption are reduced intermittently and randomly throughout the entire planning horizon as seen in the plots of e_{SYS}, no matter it is in the on-peak period or the off-peak period.

(v) Although d_{SYS} can be as low as 0 kW (i.e., all the three machines are turned off during the same time slot) in the figure, the power demand d_T of the system that counts towards the demand charge is still 75 kW as shown in Table 3. This is because it is based on the *highest* average kW measured in any 15-minute interval during the on-peak period.

(vi) Demand charges can be a significant part of the utility bill. In Table 3, the demand charge c_{DT} accounts for 39.23% of the total electricity cost c_T in the summer season. This is due to the fact that Formulation 1 does not take demand into consideration, and thus no effort is put into lowering this portion of the cost.

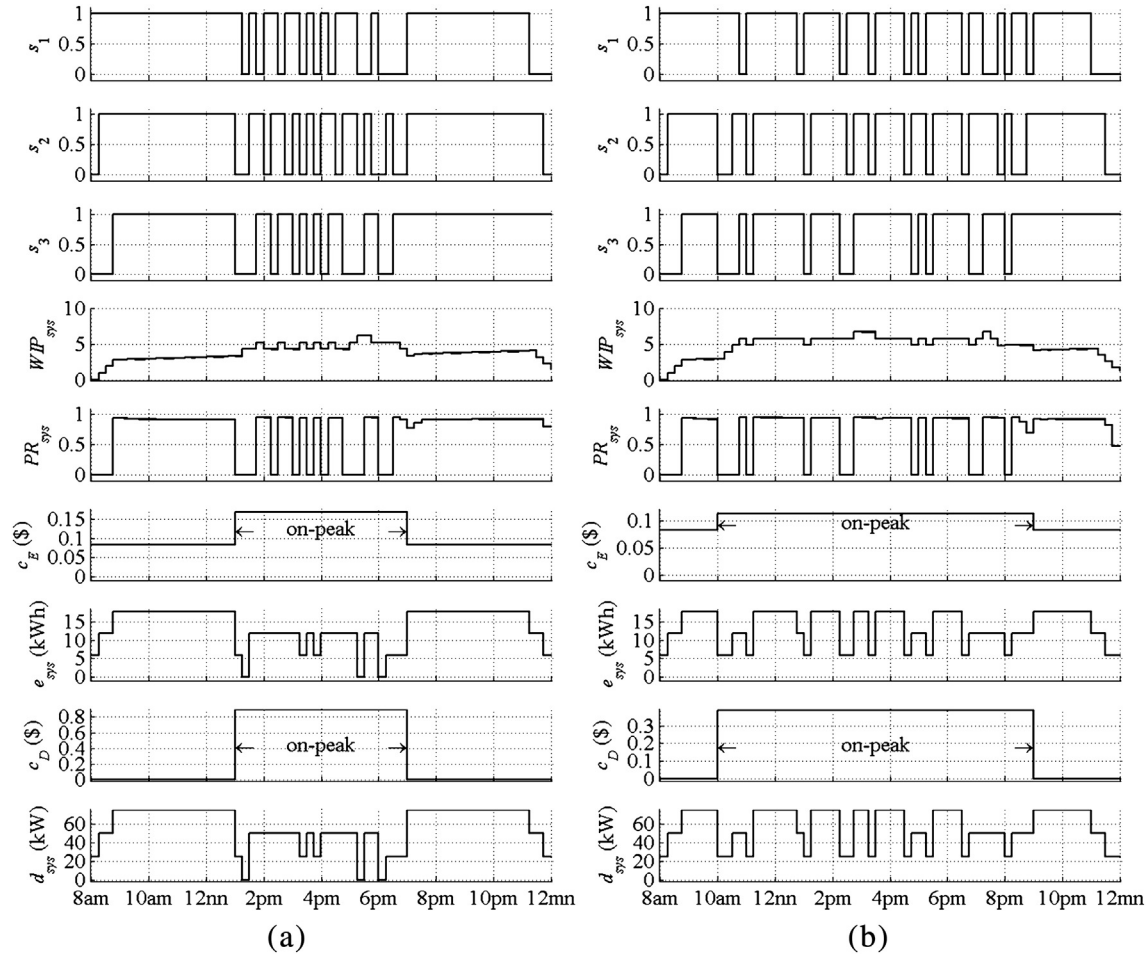


Fig. 4. Best schedules for Formulation 2 based on (a) the summer and (b) the winter TOU rates.

Table 4
Best schedules for Formulation 2.

Formulation	Solutions $s_i(t)$
Formulation 2 (summer)	$s_1(t)$ = [11111111111111111111101011011010101110010000111111111111111111000]
	$s_2(t)$ = [01111111111111111111001101101010110110100100111111111111111111110]
	$s_3(t)$ = [00011111111111111111000110110101011000110011111111111111111111111]
Formulation 2 (winter)	$s_1(t)$ = [11111111110111111101111101110111101011111011001011011111110000]
	$s_2(t)$ = [01111111001101111111011111011011110111011110100111111111100]
	$s_3(t)$ = [00011111000101111111011110011111111010111110011101111111111111]

Table 5
Objective and constraint values corresponding to the best schedules for Formulation 2.

Formulation	e_T (kWh)	d_T (kW)	c_{ET} ($)	c_{DT} ($)	c_T ($)	CP
Formulation 2 (summer)	878.750	50	93.359	44.762	138.121	45.027
Formulation 2 (winter)	872.813	75	91.655	29.000	120.655	45.038

For Formulation 2, the same procedure is followed. The best solutions together with the measures are plotted in Fig. 4. The detailed information of the best summer and winter schedules for Formulation 2 is provided in Table 4. The objective and constraint values corresponding to the schedules are listed in Table 5. Compared to Fig. 3, the results in Fig. 4 show different patterns. These results are further explained as follows:

(i) The control signals of the machines during the beginning and ending periods of the production are very similar to the ones for Formulation 1.
(ii) Other than the beginning and ending periods, the machines are controlled so that the electricity consumption is reduced only in the on-peak period when the consumption rate is higher, as seen in the plots of e_{SYS}. This is different from the solutions to Formulation 1. Evidently, Formulation 2 has the capability of tracking the TOU pricing profile.
(iii) The power demand d_T for the summer season is 50 kW, which is significantly reduced due to the avoidance of the simultaneous operation of all the three machines during the on-peak period. As a result, the demand charge c_{DT} is also reduced, as shown in Table 5.
(iv) The power demand d_T for the winter season is still 75 kW, which is the same as Formulation 1. This is because the winter on-peak period is much longer than the summer on-peak period. Due to the limited capability of the manufacturing system, it is unable to avoid the concurrent operation of all the three machines during the long on-peak period while satisfying the production constraint.
(v) For the summer season, although the total electricity consumption e_T is slightly increased compared to Formulation 1, the total electricity related cost c_T is significantly reduced. This is because the primary goal of Formulation 2 is to minimize c_T, and the consumption charge is only one component of c_T. Consequently, Formulation 2 jointly responds to both electricity consumption and power demand while Formulation 1 responds to electricity consumption only.

4.2. Insights gained from more cases

We further vary the identical machine reliability p_i' ($i = 1$, 2, and 3) from 0.91 to 0.99 with an interval of 0.02 and identical buffer capacity C_i ($i = 1$ and 2) from 1 to 9 with an interval of 2 to create 25 cases. Again, twenty trials of the binary PSO program have been run for each case. The objective values corresponding to the best solutions to these cases are provided in Figs. 5 and 6.

It should be noted that the circles in Figs. 5 and 6 mark the smallest objective function values e_T or c_T along the axis of buffer capacity C_i for each machine reliability p_i' value. For example, the 3-dimentianal coordinates (p_i', C_i, e_T) of the bottom row of Fig. 5 (a) are (0.99, 1, 872.438), (0.99, 3, 860.063), (0.99, 5, 860.063), (0.99, 7, 860.063), and (0.99, 9, 860.063). Therefore, the last four points with the same lowest e_T value 860.063 kWh are marked.

Careful observation of Figs. 5 and 6 suggests that:

(i) Since Formulation 1 responds to the electricity consumption only, the differences in the TOU rates for the summer and winter seasons have no effect on the best solutions. Therefore, Fig. 5 (a) and (b) are exactly the same.

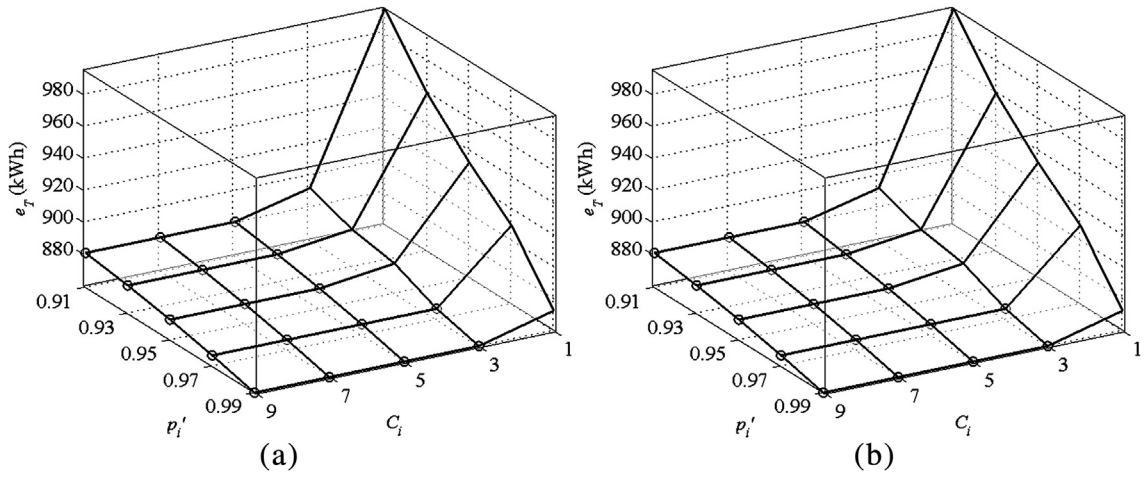


Fig. 5. Objective values corresponding to the best solutions to Formulation 1 cases based on (a) the summer and (b) the winter TOU rates.

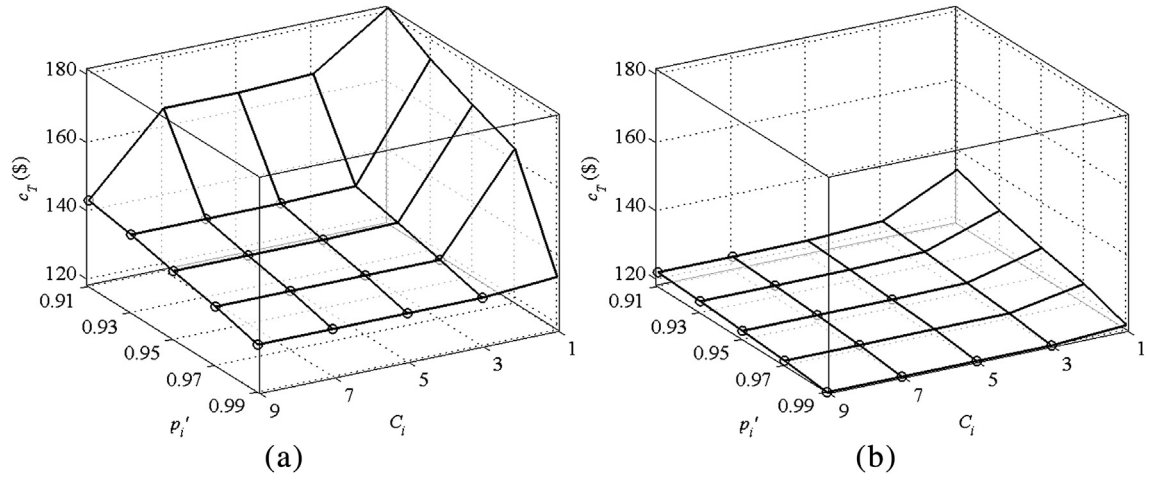


Fig. 6. Objective values corresponding to the best solutions to Formulation 2 cases based on (a) the summer and (b) the winter TOU rates.

(ii) Since Formulation 2 tracks the TOU pricing profile with regard to both electricity consumption and demand, the best solutions for the summer months shown in Fig. 6 (a) are quite different from those for the winter months in Fig. 6 (b). For every case examined, the total electricity cost c_T in the summer months is higher than that of the corresponding winter months.

(iii) As machine reliabilities p_i' increases, the electricity consumption e_T and the total electricity cost c_T corresponding to the best scheduling solutions of Formulations 1 and 2 respectively both decrease. Clearly, it is desirable to improve machine reliabilities.

(iv) As buffer capacities C_i increases, the electricity consumption e_T and the total electricity cost c_T corresponding to the best scheduling solutions of Formulations 1 and 2 respectively both decrease first, and then reach a flat zone where they both keep constant. Therefore, it is not always better to have larger buffer capacities; rather an appropriate buffer size is preferred regarding lowering e_T and c_T.

5. Conclusions

This paper investigates the TOU based electricity demand response problems aiming at minimizing the total electricity consumption and cost under the constraint of the production target. The proposed approach will fill the gaps in the literature concerning electricity demand response of manufacturing systems and help reduce the carbon footprints generated by producing a product, which can meaningfully lead to energy-efficient, demand-responsive, and cost-effective manufacturing. The binary PSO algorithm has been applied to pursuit near-optimal scheduling solutions. Although these solutions are not necessarily optimal, they can still provide remarkable insights into the nature of these problems.

As shown in Section 4, many factors may influence the scheduling solutions as well as the effectiveness and attractiveness of the TOU based demand response program to consumers. We have carefully studied the effects of factors such as problem formulations, the TOU pricing profile, machine reliabilities, and buffer capacities on the solutions. Although the results presented in Section 4 are limited to the specific examples examined, the proposed approach is a general one. For different combinations of these and many other factors, the approach can be followed to achieve acceptable near-optimal solutions on a case-by-case basis.

Acknowledgments

This work is partially supported by the U.S. National Science Foundation under Grant Number 1131537.

Appendix

A.1 Derivation of transition probabilities in equation (3)

The transition probability of buffer b_i ($i = 1, \ldots, N-1$) from state j to state j ($j = 1, ..., C_i - 1$; $C_i \geq 2$) during time slot t can be derived by the following four steps:

(i) $Q_{i,(j|j)}(t)$ is equal to the probability of {machine m_i and machine m_{i+1} each produced a part during time slot t} plus the probability of {neither machine m_i nor machine m_{i+1} produced a part during time slot t}. We analyze these two items separately.

(ii) {machine m_i and machine m_{i+1} each produced a part during time slot t} is equal to {machine m_i is up}∩{machine m_i is not starved}∩{machine m_i is not blocked}∩{machine m_{i+1} is up}∩{machine m_{i+1} is not starved}∩{machine m_{i+1} is not blocked}. Since the buffer b_i is not empty (because $j > 0$) and not full (because $j < C_i$), machine m_i is surely not blocked and machine m_{i+1} is surely not starved. Therefore the probability of {machine m_i and machine m_{i+1} each produced a part during time slot t} is equal to the probability of {machine m_i is up}∩{machine m_i is not starved}∩{machine m_{i+1} is up}∩{machine m_{i+1} is not blocked}, which is $[p_i(t) - \mathrm{ST}_i(t)][p_{i+1}(t) - \mathrm{BL}_{i+1}(t)]$.

(iii) {neither machine m_i nor machine m_{i+1} produced a part during time slot t} is equal to [{machine m_i is down}∪{machine m_i is starved}∪{machine m_i is blocked}] ∩[{machine m_{i+1} is down}∪{machine m_{i+1} is starved}∪{machine m_{i+1} is blocked}]. Since the buffer b_i is not empty (because $j > 0$) and not full (because $j < C_i$), machine m_i is surely not blocked and machine m_{i+1} is surely not starved. Therefore the probability of {neither machine m_i nor machine m_{i+1} produced a part during time slot t} is equal to the probability of [{machine m_i is down}∪{machine m_i is starved}]∩[{machine m_{i+1} is down}∪{machine m_{i+1} is blocked}], which is $[1 - p_i(t) + \mathrm{ST}_i(t)][1 - p_{i+1}(t) + \mathrm{BL}_{i+1}(t)]$.

(iv) By combining the two separate items we obtain

$$Q_{i,(j|j)}(t) = [p_i(t) - ST_i(t)][p_{i+1}(t) - BL_{i+1}(t)] + [1 - p_i(t) + ST_i(t)][1 - p_{i+1}(t) + BL_{i+1}(t)], \quad (j = 1, \ldots, C_i - 1;\ C_i \geq 2)$$

The other transition probabilities in (3) can be derived similarly following the above logic. The derivations are omitted here.

A.2 Derivation of starvation probabilities in equation (7)

Since the first machine is never starved, the probability of starvation of machine m_1 during time slot t is 0. The probability of starvation of machine m_i ($i = 2, \ldots, N$) during time slot t is equal to the probability of {machine m_i is up}∩{buffer b_{i-1} is empty at the end of time slot $t-1$}, which is $p_i(t)q_{i-1,0}(t-1)$. This concludes the derivation.

A.3 Derivation of blockage probabilities in equation (8)

Since the last machine is never blocked, the probability of blockage of machine m_N during time slot t is 0. The probability of blockage of machine m_i ($i = 1, \ldots, N-1$) during time slot t is equal to the probability of {machine m_i is up}∩{buffer b_i is full at the end of time slot $t-1$}∩{machine m_{i+1} did not produce any part during time slot t}. The last item is further analyzed as follows. {machine m_{i+1} did not produce any part during time slot t} is equal to {machine m_{i+1} is down}∪{machine m_{i+1} is starved}∪{machine m_{i+1} is blocked}. Since {buffer b_i is full at the end of time slot $t-1$} makes {machine m_{i+1} is starved} impossible, the probability of blockage of machine m_i ($i = 1, \ldots, N-1$) during time slot t is equal to the probability of {machine m_i is up}∩{buffer b_i is full at the end of time slot $t-1$}∩[{machine m_{i+1} is down}∪{machine m_{i+1} is blocked}], which is $p_i(t)q_{i,C_i}(t-1)[1 - p_{i+1}(t) + \mathrm{BL}_{i+1}(t)]$. This concludes the derivation.

References

[1] U.S. Energy Information Administration. Annual energy review 2011. Retrieved January 15, 2013, from, http://www.eia.gov/totalenergy/data/annual/pdf/aer.pdf; 2012.

[2] U.S. Environmental Protection Agency. Sources of greenhouse gas emissions. Retrieved January 15, 2013, from, http://www.epa.gov/climatechange/ghgemissions/sources/electricity.html; 2011.

[3] U.S. Energy Information Administration. Electric power annual 2010. Retrieved January 15, 2013, from, http://www.eia.gov/electricity/annual/pdf/epa.pdf; 2010.

[4] ComEd. Load forecast for five-year planning period (June 2013 – May 2018). Retrieved January 15, 2013, from, http://www2.illinois.gov/ipa/Documents/AppendixII-ComEdLoadForecast.pdf; 2012.

[5] Chupka MW, Earle R, Fox-Penner P, Hledik R. Transforming America's power industry: the investment challenge 2010–2030. Retrieved January 15, 2013, from, http://www.brattle.com/_documents/uploadlibrary/upload725.pdf; 2008.

[6] U.S. Environmental Protection Agency. Greenhouse gas equivalencies calculator: calculations and references. Retrieved January 15, 2013, from, http://www.epa.gov/cleanenergy/energy-resources/refs.html; 2012.

[7] U.S. Department of Energy. President Obama sets a target for cutting U.S. Greenhouse gas emissions. Retrieved January 15, 2013, from, http://apps1.eere.energy.gov/news/news_detail.cfm/news_id=15650; 2009.

[8] U.S. Federal Energy Regulatory Commission. Assessment of demand response and advanced metering staff report 2012. Retrieved January 15, 2013, from, http://www.ferc.gov/legal/staff-reports/12-20-12-demand-response.pdf; 2012.

[9] Finn P, Fitzpatrick C, Connolly D, Leahy M, Relihan L. Facilitation of renewable electricity using price based appliance control in Ireland's electricity market. Energy 2011;36(5):2952–60.

[10] Pina A, Silva C, Ferrao P. The impact of demand side management strategies in the penetration of renewable electricity. Energy 2012;41(1):128–37.

[11] Quiggin D, Cornell S, Tierney M, Buswell R. A simulation and optimisation study: towards a decentralised microgrid, using real world fluctuation data. Energy 2012;41(1):549–59.

[12] Walawalkar R, Fernands S, Thakur N, Chevva KR. Evolution and current status of demand response (DR) in electricity markets: Insights from PJM and NYISO. Energy 2010;35(4):1553–60.

[13] 109th U.S. Congress. Energy Policy Act of 2005. Retrieved January 11, 2013, from, http://www.govtrack.us/congress/bills/109/hr6; 2005.

[14] Orange and Rockland Utilities. Service classification No. 20. Retrieved January 15, 2013, from, https://www.oru.com/documents/tariffsandregulatorydocuments/ny/electrictariff/electricsc20.pdf; 2012.

[15] Nikzad M, Mozafari B, Bashirvand M, Solaymani S, Ranjbar AM. Designing time-of-use program based on stochastic security constrained unit commitment considering reliability index. Energy 2012;41(1):541–8.

[16] Cook B, Gazzano J, Gunay Z, Hiller L, Mahajan S, Taskan A, et al. The smart meter and a smarter consumer: quantifying the benefits of smart meter implementation in the United States. Chem Cent J 2012;6(S1).

[17] Lima CAF, Navas JRP. Smart metering and systems to support a conscious use of water and electricity. Energy 2012;45(1):528–40.

[18] Gungor VC, Sahin D, Kocak T, Ergut S, Buccella C, Cecati C, et al. A survey on smart grid potential applications and communication requirements. IEEE Trans Ind Inform 2013;9(1):28–42.

[19] Finn P, Fitzpatrick C, Connolly D. Demand side management of electric car charging: benefits for consumer and grid. Energy 2012;42(1):358–63.

[20] Torriti J. Price-based demand side management: assessing the impacts of time-of-use tariffs on residential electricity demand and peak shifting in Northern Italy. Energy 2012;44(1):576–83.

[21] Herter K, Wayland S. Residential response to critical-peak pricing of electricity: California evidence. Energy 2010;35(4):1561–7.

[22] Rastegar M, Fotuhi-Firuzabad M, Aminifar F. Load commitment in a smart home. Appl Energy 2012;96:45–54.

[23] van Ruijven B, de Vries B, van Vuuren DP, van der Sluijs JP. A global model for residential energy use: uncertainty in calibration to regional data. Energy 2010;35(1):269–82.

[24] Venkatesan N, Solanki J, Solanki SK. Residential demand response model and impact on voltage profile and losses of an electric distribution network. Appl Energy 2012;96:84–91.

[25] Corno F, Razzak F. Intelligent energy optimization for user intelligible goals in smart home environments. IEEE Trans Smart Grid 2012;3(4):2128–35.

[26] Houwing M, Negenborn RR, De Schutter B. Demand response with micro-CHP systems. Proc IEEE 2011;99(1):200–13.

[27] Liang Y, Levine DI, Shen Z-J. Thermostats for the smart grid: models, benchmarks, and insights. Energy J 2012;33(4):61–95.

[28] Motegi N, Piette MA, Watson DS, Kiliccote S, Xu P. Introduction to commercial building control strategies and techniques for demand response. Retrieved January 15, 2013, from, http://gaia.lbl.gov/btech/papers/59975.pdf; 2007.

[29] Wang L, Wang Z, Yang R. Intelligent multiagent control system for energy and comfort management in smart and sustainable buildings. IEEE Trans Smart Grid 2012;3(2):605–17.

[30] Chao X, Chen FY. An optimal production and shutdown strategy when a supplier offers an incentive program. Manuf Serv Operat Manag 2005;7(2):130–43.

[31] Lewis G. Strategies to increase California food processing industry demand response participation: a scoping study. Retrieved January 15, 2013, from, http://drrc.lbl.gov/system/files/lbnl-63668.pdf; 2007.

[32] Logenthiran T, Srinivasan D, Tan Zong S. Demand side management in smart grid using heuristic optimization. IEEE Trans Smart Grid 2012;3(3):1244–52.

[33] McKane A, Rhyne I, Lekov A, Thompson L, Piette MA. Automated demand response: the missing link in the electricity value chain. Retrieved January 15, 2013, from, http://drrc.lbl.gov/sites/drrc.lbl.gov/files/lbnl-2736e.pdf; 2008.

[34] Gershwin SB. Manufacturing systems engineering. Prentice-Hall; 1994.

[35] Li J, Meerkov SM. Production systems engineering. Springer; 2008.

[36] Li J, Meerkov SM, Zhang L. Production systems engineering: problems, solutions, and applications. Annu Rev Control 2010;34(1):73–88.

[37] Li L. Bottleneck detection of complex manufacturing systems using a data-driven method. Int J Prod Res 2009;47(24):6929–40.

[38] Li L, Chang Q, Ni J, Biller S. Real time production improvement through bottleneck control. Int J Prod Res 2009;47(21):6145–58.

[39] Gershwin SB, Werner LM. An approximate analytical method for evaluating the performance of closed-loop flow systems with unreliable machines and finite buffers. Int J Prod Res 2007;45(14):3085–111.

[40] Colledani M, Gershwin S. A decomposition method for approximate evaluation of continuous flow multi-stage lines with general Markovian machines. Ann Operat Res 2011:1–36.

[41] Li J, Blumenfeld DE, Huang N, Alden JM. Throughput analysis of production systems: recent advances and future topics. Int J Prod Res 2009;47(14):3823–51.

[42] Meerkov SM, Zhang L. Transient behavior of serial production lines with Bernoulli machines. IIE Trans 2008;40(3):297–312.

[43] Meerkov SM, Shimkin N, Zhang L. Transient behavior of two-machine geometric production lines. IEEE Trans Autom Control 2010;55(2):453–8.

[44] Ashok S. Peak-load management in steel plants. Appl Energy 2006;83(5):413–24.

[45] Fang K, Uhan N, Zhao F, Sutherland JW. A new approach to scheduling in manufacturing for power consumption and carbon footprint reduction. J Manuf Syst 2011;30(4):234–40.

[46] Faria P, Vale Z, Soares J, Ferreira J. Demand response management in power systems using a particle swarm optimization approach. IEEE Intell Syst 2011;PP(99):1–9.

[47] Soares J, Silva M, Sousa T, Vale Z, Morais H. Distributed energy resource short-term scheduling using signaled particle swarm optimization. Energy 2012;42(1):466–76.

[48] Wang Y, Li L, Ni J, Huang S. Form tolerance evaluation of toroidal surfaces using particle swarm optimization. J Manuf Sci Eng Trans ASME 2009;131(5):051015.1–9.
[49] Wang Y, Li L. Heterogeneous redundancy allocation for series-parallel multi-state systems using hybrid particle swarm optimization and local search. IEEE Trans Syst Man Cybern Part A Syst Hum 2012;42(2):464–74.
[50] Thangaraj R, Pant M, Abraham A, Bouvry P. Particle swarm optimization: hybridization perspectives and experimental illustrations. Appl Math Comput 2011;217(12):5208–26.
[51] Poli R, Kennedy J, Blackwell T. Particle swarm optimization: an overview. Swarm Intell 2007;1(1):33–57.
[52] Roy R, Hinduja S, Teti R. Recent advances in engineering design optimisation: challenges and future trends. Cirp Ann Manuf Technol 2008;57(2):697–715.
[53] Martens D, Baesens B, Fawcett T. Editorial survey: swarm intelligence for data mining. Mach Learn 2011;82(1):1–42.
[54] Kennedy J, Eberhart RC. A discrete binary version of the particle swarm algorithm. In: 1997 IEEE International Conference on Systems, Man, and Cybernetics, Orlando, FL, USA, vol. 5; 1997. p. 4104–8.
[55] Rezaee Jordehi A, Jasni J. Parameter selection in particle swarm optimisation: a survey. J Exp Theor Artif Intell 2013:1–16.
[56] Trelea IC. The particle swarm optimization algorithm: convergence analysis and parameter selection. Inf Process Lett 2003;85(6):317–25.
[57] Clerc M, Kennedy J. The particle swarm – explosion, stability, and convergence in a multidimensional complex space. IEEE Trans Evol Comput 2002;6(1):58–73.
[58] Shi Y, Eberhart RC. Parameter selection in particle swarm optimization. In: Porto VW, Saravanan N, Waagen D, Eiben AE, editors. Evolutionary Programming VII. Berlin Heidelberg: Springer; 1998. p. 591–600.
[59] Clerc M. Binary particle swarm optimisers: toolbox, derivations, and mathematical insights. Retrieved January 15, 2013, from, http://hal.archives-ouvertes.fr/docs/00/12/28/09/PDF/Binary_PSO.pdf; 2007.